

ORBIT

SaaS Project Management Platform

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React 19 Node.js MongoDB Socket.io Redux

1 Project Overview

Orbit is a production-grade, full-stack SaaS project management platform built for teams. It provides a Kanban-style workflow management system with real-time collaboration, multi-organization support, and role-based access control.

The project was built over 3+ months as a portfolio project to demonstrate full-stack engineering capabilities including system design, API design, real-time communication, authentication, authorization, and production deployment.

1.1 Motivation

Modern project management tools like Jira, Linear, and ClickUp share a common architecture:

- Multi-tenant organizations
- Role-based permissions
- Real-time collaboration
- Kanban / board views

Orbit replicates this architecture from scratch, making deliberate engineering decisions at every layer.

1.2 Key Metrics

Metric	Value
Total pages	15+
API endpoints	40+
Database models	11
Real-time events	8
Development time	3+ months

2 Live Deployment

Resource	URL / Path	Status
Frontend	orbit-frontend-app.vercel.app	■ Permanent
Backend API	orbit-backend-production-76bf.up.railway.app/api	■ ~30 days
API Health	.../api/health	■ Live
API Docs	.../api/docs	■ Live
Frontend Repo	github.com/AhmadAbdullah75/orbit-frontend	■ Public
Backend Repo	github.com/AhmadAbdullah75/orbit-backend	■ Public

Deployment Lifespan Notes

- Vercel (Frontend): Free tier — permanent, no expiry.
- MongoDB Atlas (Database): Free M0 tier — permanent.
- Railway (Backend): \$5 free credit — approximately 30 days. Will be migrated after credit is consumed.

3 Feature Specification

3.1 Authentication

Feature	Description
Email registration	Name, email, password with bcrypt hashing
Email/password login	JWT token returned, 30-day expiry
Google OAuth	Passport.js Google Strategy, account merging
Forgot password	Resend email with 24-hour reset link
Reset password	Token-validated password update
Auto-logout	401 interceptor clears session
Remember last org	localStorage persists last workspace

3.2 Organizations

- Create unlimited organizations; switch between them instantly
- Each organization has fully isolated data
- Invite members via email with role assignment
- Invitation tokens expire after 30 days
- Real-time invitation delivery via Socket.io

3.3 Kanban Board

- Drag-and-drop tasks between any columns (TouchSensor for mobile — 250 ms activation)
- Create, edit, and delete tasks
- Subtasks with completion tracking and progress bars
- Rich text descriptions via Tiptap
- File attachments via Cloudinary
- Assignee selection from org members
- Priority levels: Urgent, High, Medium, Low
- Due dates with overdue indicators
- Labels / tags
- Column management: add, rename, delete; unique column name enforcement

3.4 Task Management

- Global tasks view across all projects
- Filter by project, priority, status, assignee; search by title
- "Assigned to Me" / "All Tasks" toggle
- Task detail panel with comments

3.5 Analytics

- Overall workspace completion rate (donut chart)
- Per-project completion progress bars
- Priority distribution breakdown
- Contributor output: assigned vs completed per member
- Real-time refresh with tab visibility detection

3.6 Notifications

- Bell icon with unread count badge
- Real-time delivery via Socket.io — no page refresh needed

- Invitation notifications with Accept / Decline modal
- "Already joined" detection for old notifications
- Mark as read / mark all as read
- Auto-poll every 120 seconds as fallback

3.7 Activity Log

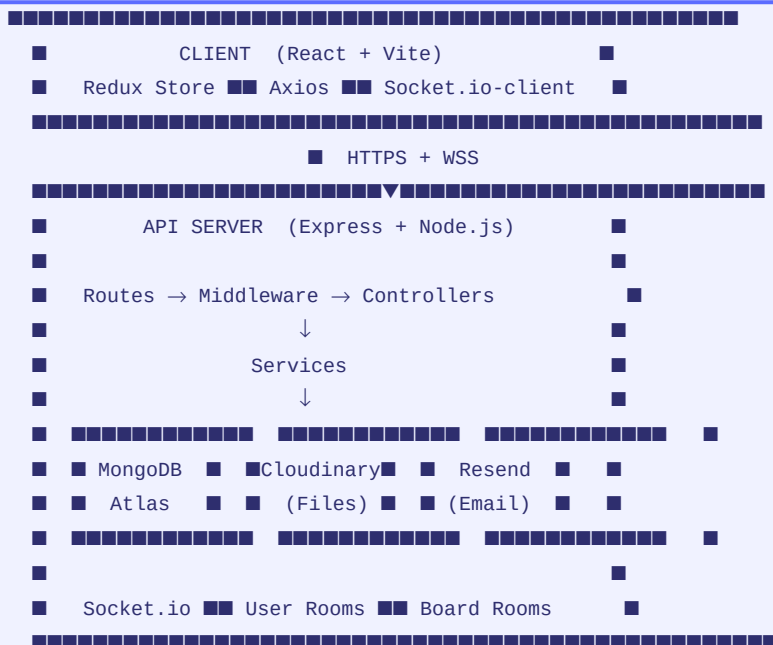
- Full audit trail for all organization events
- Infinite scroll pagination
- Filter by entity type and action
- Bold formatting for task / project names
- Relative timestamps

3.8 Settings

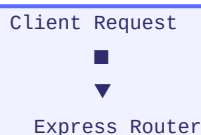
- Profile picture upload via Cloudinary (max 5 MB)
- Display name update; email is read-only to prevent hijacking
- Workspace name update (admin / owner only)
- Notification preferences stored in localStorage
- Danger Zone: delete organization (owner only)
- Upgrade section (coming-soon modal)

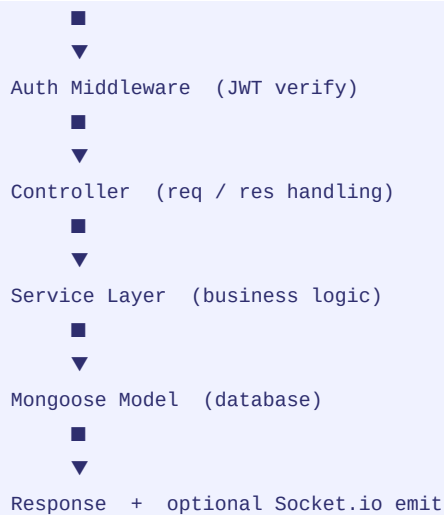
4 System Architecture

4.1 High-Level Overview

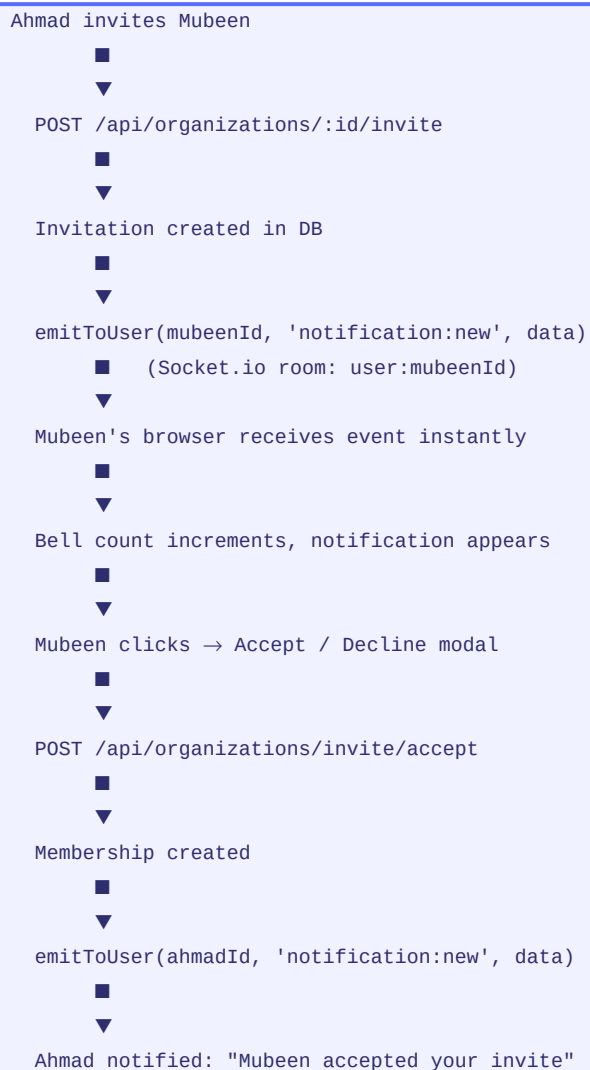


4.2 Request Flow





4.3 Real-time Invitation Flow



5 Tech Stack

5.1 Frontend

Package	Version	Purpose
react	19	UI framework
vite	7	Build tool
@reduxjs/toolkit	2	State management
redux-persist	6	State persistence
react-router-dom	6	Client routing
axios	1	HTTP client
socket.io-client	4	Real-time
@dnd-kit/core	6	Drag and drop
@dnd-kit/sortable	8	Sortable lists
framer-motion	11	Animations
@tiptap/react	2	Rich text editor

5.2 Backend

Package	Version	Purpose
express	4	HTTP framework
mongoose	8	MongoDB ODM
jsonwebtoken	9	JWT tokens
bcryptjs	2	Password hashing
passport	0.7	OAuth strategies
passport-google-oauth20	2	Google OAuth
socket.io	4	Real-time server
multer	1	File upload
cloudinary	2	File storage
resend	4	Email delivery
swagger-ui-express	5	API documentation
cors	2	Cross-origin
cookie-parser	1	HTTP cookies

6 Database Design

6.1 Schema Overview

User

```

    ■■■ _id, name, email, password (hashed)
    ■■■ avatar (Cloudinary URL)
    ■■■ googleId (for OAuth)
    ■■■ isVerified
    ■■■ passwordResetToken, passwordResetExpires

```

Organization

```

    ■■■ _id, name, description
    ■■■ owner (ref: User)
    ■■■ createdAt

```

```

Membership (junction: User <-> Organization)

```

```

    ■■■ user (ref: User)
    ■■■ organization (ref: Organization)
    ■■■ role: 'owner' | 'admin' | 'member'
    ■■■ joinedAt

Invitation
    ■■■ organization (ref: Organization)
    ■■■ email (invitee email)
    ■■■ role
    ■■■ token (unique, plain text)
    ■■■ status: 'pending' | 'accepted' | 'expired' | 'revoked'
    ■■■ invitedBy (ref: User)
    ■■■ expiresAt (30 days)

Project
    ■■■ organization (ref: Organization)
    ■■■ name, description, color
    ■■■ status: 'active' | 'completed' | 'archived'
    ■■■ members[] (ref: User)

Board
    ■■■ project (ref: Project)
    ■■■ name

Column
    ■■■ board (ref: Board)
    ■■■ name
    ■■■ order

Task
    ■■■ column (ref: Column)
    ■■■ title, description (HTML from Tiptap)
    ■■■ priority: 'urgent' | 'high' | 'medium' | 'low'
    ■■■ assignees[] (ref: User)
    ■■■ labels[], startDate, dueDate, attachments[]
    ■■■ subtasks[] (embedded) { title, completed }
    ■■■ order

Comment
    ■■■ task (ref: Task)
    ■■■ user (ref: User)
    ■■■ content (HTML)

Activity
    ■■■ organization (ref: Organization)
    ■■■ user (ref: User)
    ■■■ action (string)
    ■■■ entity: 'task' | 'project' | 'member' | 'column'
    ■■■ metadata

Notification
    ■■■ recipient (ref: User)
    ■■■ sender (ref: User)
    ■■■ type: 'invitation' | 'info' | 'task'
    ■■■ message
    ■■■ metadata (orgId, role, token, etc.)
```

■■■ isRead

7 API Reference

7.1 Base URL

`https://orbit-backend-production-76bf.up.railway.app/api`

7.2 Authentication Header

`Authorization: Bearer <JWT_TOKEN>`

7.3 Standard Response Format

```
{
  "status": "success" | "error" | "fail",
  "data": { ... },
  "message": "Human readable message"
}
```

7.4 Auth Endpoints

POST	/auth/register	body: { name, email, password }	→ { token, user }
POST	/auth/login	body: { email, password }	→ { token, user }
PATCH	/auth/profile	body: { name, avatar }	→ { user }
POST	/auth/forgot-password	body: { email }	→ { emailSent, resetUrl, message }
POST	/auth/reset-password	body: { token, password }	→ { message }

7.5 Organization Endpoints

GET	/organizations	→ { organizations[] }
POST	/organizations	body: { name } → { organization }
POST	/organizations/:id/invite	body: { email, role } → { invitation }
POST	/organizations/invite/accept	body: { token } → { membership }
GET	/organizations/:id/all-tasks	→ { tasks[], total, open, done }

7.6 Task Endpoints

GET	/tasks?boardId=:id	→ { tasks[] }
POST	/boards/:bid/columns/:cid/tasks	→ { task }
PATCH	/tasks/:id/move	body: { columnId } → { task }

8 Authentication & Security

8.1 JWT Strategy

- Access token: 30-day expiry
- Stored in localStorage
- Sent as Authorization: Bearer <token>
- Auto-logout on 401 response

8.2 Password Security

- bcrypt with salt rounds: 12
- Password never returned in API responses
- Reset tokens hashed with SHA-256 before storage

8.3 Google OAuth Flow

```

User clicks "Continue with Google"
  ▼
Redirect to Google consent screen
  ▼
Google redirects to /api/auth/google/callback
  ▼
Passport verifies, creates / updates User
  ▼
Redirect to CLIENT_URL/auth/google/success?token=...
  ▼
Frontend stores token, navigates to dashboard
  
```

8.4 CORS Configuration

```

origin:      process.env.CLIENT_URL
credentials: true
methods:     ['GET', 'POST', 'PUT', 'PATCH', 'DELETE']
  
```

9 Real-time System

9.1 Socket.io Rooms

```

Board rooms:  board:{boardId}
User rooms:   user:{userId}
  
```

9.2 Client Connection

```

const socket = io(VITE_SOCKET_URL, {
  auth: { token },
  transports: ['websocket', 'polling']
})

// Join personal room on connect
socket.emit('join:user', userId)
  
```

9.3 Server Helper

```

// Emit to specific user
export const emitToUser = (userId, event, data) => {
  global.io.to(`user:${userId}`).emit(event, data)
}
  
```

10 Role-Based Access Control

10.1 Permission Matrix

Permission	Owner	Admin	Member
Delete organization	■	■	■
Rename organization	■	■	■
Transfer ownership	■	■	■
Invite members	■	■	■
Remove members	■	■	■
Change member roles	■	■*	■
Create projects	■	■	■
Delete projects	■	■	■
Manage board	■	■	■
Create tasks	■	■	■
Update tasks	■	■	■
View analytics	■	■	■

* Admin cannot change own role or the owner's role.

10.2 Permission Enforcement

Permissions are enforced at two layers:

- Frontend — UI elements are hidden or shown based on the user's role.
- Backend — the service layer validates the role before any database operation.

11 Frontend Architecture

11.1 Redux State Shape

```

Redux Store
  ■■■ auth
  ■   ■■■ user (name, email, avatar)
  ■   ■■■ token
  ■   ■■■ activeOrgId
  ■■■ (persisted to localStorage via redux-persist)
  
```

11.2 Axios API Layer

- Automatically attaches Bearer token to every request
- 401 response → auto logout + redirect to login
- Network errors surfaced as user-friendly messages via `userMessage` property

11.3 useAutoRefresh Hook

- Polls data at a specified interval
- Pauses automatically when the browser tab is hidden
- Resumes immediately when the tab becomes active again

- Cleans up interval on component unmount

11.4 Mobile Architecture

- Desktop: fixed left sidebar — 240 px wide / 64 px when collapsed
- Mobile: hamburger button → full drawer (280 px) with backdrop
- Mobile nav: 5-item bottom navigation bar
- Touch drag: @dnd-kit TouchSensor with 250 ms activation delay

12 Deployment Guide

12.1 Frontend — Vercel

```
# 1. Push to GitHub
git push origin main

# 2. Connect repo to Vercel
#   vercel.com → New Project → Import GitHub

# 3. Add environment variables in Vercel dashboard
VITE_API_URL=https://your-backend.railway.app/api
VITE_SOCKET_URL=https://your-backend.railway.app

# 4. vercel.json (required for React Router)
{
  "rewrites": [{ "source": "/(.*)", "destination": "/index.html" }]
}
```

12.2 Backend — Railway

```
# 1. Push to GitHub
git push origin main

# 2. Connect repo to Railway
#   railway.app → New Project → Deploy from GitHub

# 3. Add all environment variables in Railway dashboard
# 4. Generate domain: Railway → Settings → Networking

# Dockerfile (included in repo):
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --only=production
COPY . .
EXPOSE 5000
ENV NODE_ENV=production
CMD ["node", "src/server.js"]
```

12.3 MongoDB Atlas

- cloud.mongodb.com → Create free M0 cluster
- Database Access → Create user with read/write permissions
- Network Access → Add 0.0.0.0/0 (allow all IPs)

- Connect → Drivers → Copy connection string
- Replace <password> and append your database name

12.4 Google OAuth Setup

- console.cloud.google.com → Create project → Enable Google OAuth
- Credentials → Create OAuth Client (Web Application)
- Authorized redirect URIs: https://your-backend.railway.app/api/auth/google/callback
- Copy Client ID and Client Secret to .env

13 Environment Variables

13.1 Backend (.env)

Variable	Description	Example
NODE_ENV	Environment	production
PORT	Server port	5000
MONGODB_URI	MongoDB connection	mongodb+srv://...
JWT_SECRET	JWT signing key	random_string_32+
JWT_EXPIRES_IN	Token expiry	30d
CLIENT_URL	Frontend URL	https://orbit-frontend-app.vercel.app
GOOGLE_CLIENT_ID	OAuth client ID	915651...apps.googleusercontent.com
GOOGLE_CLIENT_SECRET	OAuth secret	GOCSPX-...
GOOGLE_CALLBACK_URL	OAuth redirect	https://backend.railway.app/api/auth/...
CLOUDINARY_CLOUD_NAME	Cloud name	your_cloud
CLOUDINARY_API_KEY	API key	397816...
CLOUDINARY_API_SECRET	API secret	aRv4Kb...
RESEND_API_KEY	Email API key	re_...
FROM_EMAIL	Sender email	onboarding@resend.dev

13.2 Frontend (.env)

Variable	Description	Example
VITE_API_URL	Backend API URL	https://backend.railway.app/api
VITE_SOCKET_URL	Socket.io URL	https://backend.railway.app

14 Known Limitations

Limitation	Reason	Workaround
Backend 30-day lifespan	Railway \$5 free credit	Redeploy on alternative host
No email verification	Demo purposes	isVerified auto-set to true
Resend accepts fake emails	API returns 200 always	Reset URL returned in response
No file size limit on board	Cloudinary free tier	10 MB limit enforced in upload
Google OAuth needs hard refresh	Font / cache issue	Fixed with npm font bundle

A Local Development Setup

```
# Terminal 1 – Backend
cd orbit-backend
cp .env.example .env    # fill in .env values
npm install
npm run dev              # http://localhost:5000

# Terminal 2 – Frontend
cd orbit-frontend
cp .env.example .env
# VITE_API_URL=http://localhost:5000/api
# VITE_SOCKET_URL=http://localhost:5000
npm install
npm run dev              # http://localhost:5173
```

B Testing Checklist

- Register with email → redirects to dashboard
- Login with email → opens correct organization
- Google Sign-In → creates / logs in account
- Forgot password (real email) → email received
- Forgot password (fake email) → reset link displayed on screen
- Create organization → appears in switcher
- Invite member → real-time notification delivery
- Accept invitation → joins org, inviter notified
- Create project → appears in list
- Open board → columns visible
- Create task → appears in correct column
- Drag task → moves between columns
- Upload avatar → persists after logout / login
- Dark / light theme → persists across sessions
- Mobile navigation → bottom nav works
- Mobile drag-drop → hold 250 ms then drag

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